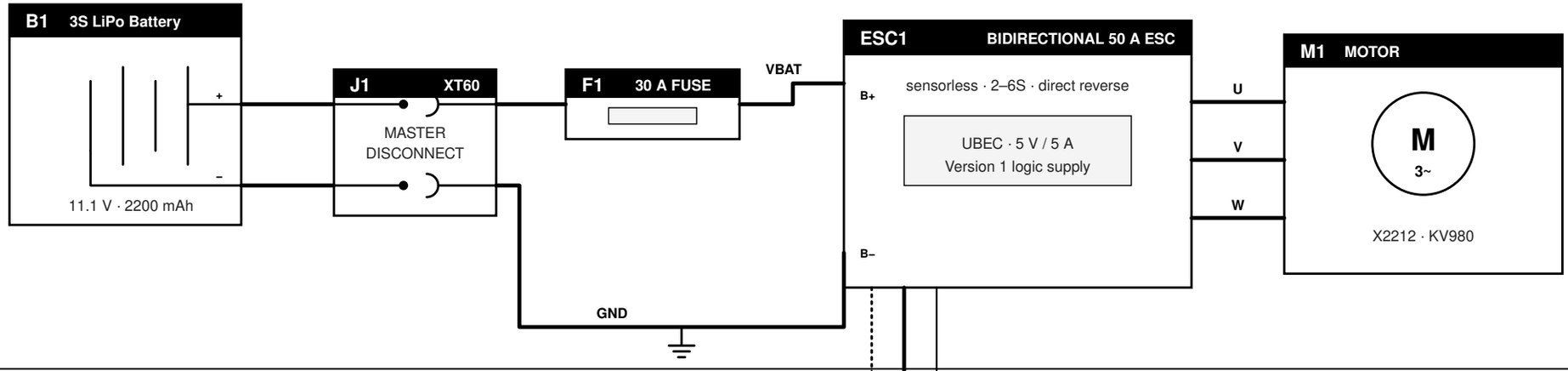
DKS-BASIC ESP32 RC CAR — ELECTRICAL SYSTEM

Version 1 · PS4 Bluetooth control · 3S LiPo · sensorless brushless drive

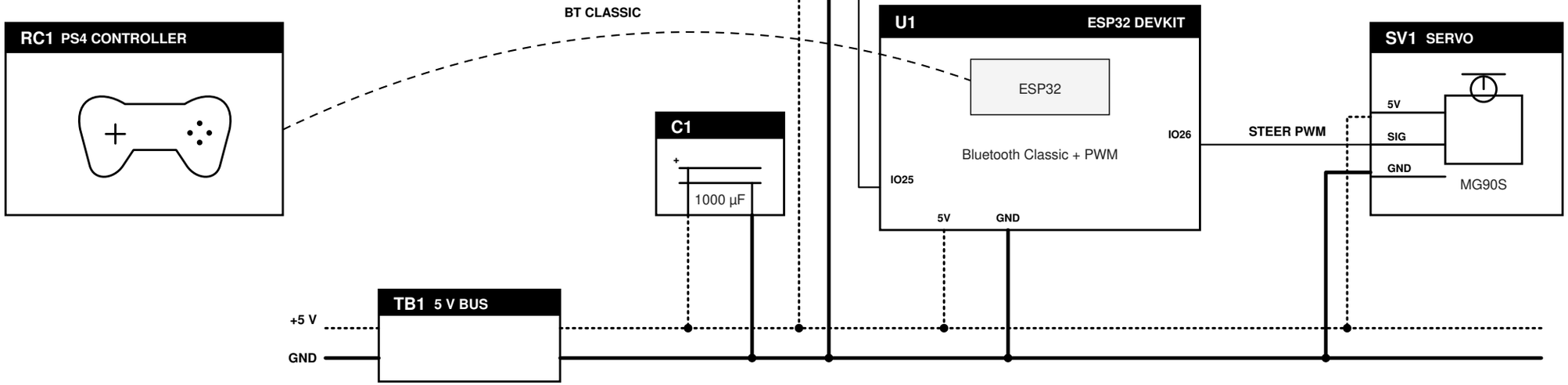
SHEET 1 / 1

A3 · NTS

A · POWERTRAIN



B · CONTROL & LOW VOLTAGE



WIRE KEY

- Power & ground
- +5 V logic rail
- Signal (PWM)
- Wireless link

SAFETY & START-UP

1. J1 (XT60) is the master disconnect; there is no separate switch. Break it only with the motor stopped and throttle neutral. A small inrush spark is normal.
2. F1 is over-current protection, not an on/off switch. Keep it close to B1 positive and run 14 AWG from F1 straight to ESC1 B+.
3. Disconnect B1 before wiring or USB. Never connect the 3S rail to U1, SV1 or the 5 V bus.
4. Firmware holds neutral (~1500 µs) until ESC1 is armed and on link loss. Commission wheels-up; to reverse, disconnect B1 and swap two phases.

PROJECT	DRAWING	DOC	REV	DATE	STATUS
DKS-Basic ESP32 RC Car	Electrical System	RC-CAR-ELEC-V1	1.2	2026-08-31	RELEASE